REQUEST FOR INFORMATION (RFI)

Department of Homeland Security (DHS)
Science and Technology (S&T) Directorate
Office of Mission and Capability Support (MCS)
Advanced Communications Experiment – Cross-border Autonomous Vehicle Session Persistence Experiment and Research

1.0 General Information

1.1 Introduction

The U.S. Department of Homeland Security (DHS), Office of Procurement Operations (OPO) issues this Request for Information (RFI) on behalf of the DHS Science and Technology Directorate (S&T), Office of Mission and Capability Support (MCS).

NOTE: THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for market research, information, and planning purposes; it does not constitute a solicitation or any commitment or intent to issue a solicitation in the future. DHS intends to use this RFI to seek participants in a multi-day experiment along the U.S.-Canada border in late fall 2026. This RFI does not commit DHS to contract for any supply or service. DHS is not seeking proposals. Respondents are advised that DHS will not pay for any costs incurred in responding to this RFI or for participating in scheduled demonstration events. All costs associated with responding to this RFI and participating in future experiments will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future solicitation if one is issued. The information provided in this RFI is subject to change and is not binding on DHS.

1.2 Program Information

a. Program Name: Critical Infrastructure Security Resilience Research

b. Managing Program Office: Office of Mission and Capability Support (MCS)

1.3 Procurement Information

a. 2023 North American Industry Classification System (NAICS) codes:

- 541712 – Research and Development in the Physical, Engineering, and Life Sciences (except Nanotechnology and Biotechnology)

b. Product Service Codes:

- 1550 – "Unmanned Aircraft"

2.0 Background

2.1 Introduction

The U.S. Department of Homeland Security (DHS) Science and Technology Directorate's (S&T), in collaboration with Defence Research and Development Canada, Centre for Security Science (DRDC CSS), is hosting the 2026 Advanced Communications Experiment Cross-border Autonomous Vehicle Session Persistence Experiment and Research (ACE-CASPER). This multi-day experiment will take place along the U.S.-Canada border and will simulate a national emergency response scenario, evaluating how autonomous vehicles (AV), unmanned aerial systems (UAS), and 5G-enabled technologies can support persistent cross-border communications, enhance situational awareness, and improve coordination between U.S. and Canadian response teams. During the experiment, UAS and AV will move across both sides of the border streaming live video to a bi-national command and control center (C2), deploy sensor-based payloads to assess the situation, relay that data back to C2, and use that data to inform decision making and response. The experiment will demonstrate:

- Secure, resilient 5G session persistence, with connectivity across U.S. and Canadian border networks;
- Seamless command and control (C2) handoff, and real-time data interoperability;
- Operational readiness validation for DHS components and DRDC with State, Provincial, and Local players in cross-border coordination; and
- Bi-national response coordination using enhanced, real-time situational awareness tools.

Vehicle autonomy level is secondary to the experiment's primary objective of demonstrating resilient, persistent 5G communications across U.S. and Canadian networks.

2.2 Objective

The ACE-CASPER Experiment aims to evaluate the operational readiness and technological capabilities of 5G Session Persistence and 5G connectivity with UAS and AV systems in a cross-border emergency scenario. Vendors are expected to demonstrate solutions that ensure secure, resilient 5G connectivity, real-time situational awareness, and seamless interoperability between U.S. and Canadian networks.

The experiment does not require autonomous decision-making or AI-driven navigation; the primary focus is validating reliable 5G connectivity, session persistence, and command-and-control performance.

<u>Please see the details in the attached ACE-CASPER Experiment 2026 for UAS and AV operational capabilities, participation requirements, and submission instructions.</u>

3.0 Responses to this RFI

Information or inquiries submitted in response to this RFI must be submitted via email to **NGFR@hq.dhs.gov** by or before **May 8, 2026, 5:00 p.m. EST**. Submissions must complete the required information request in the attached RFI document and may include an optional 5-page whitepaper.

DO NOT SUBMIT ACQUISITION PROPOSALS. Submit participation applications. No financial contracts will be awarded based on this announcement, but selected participants may be asked to sign a Limited Purpose Cooperative Research and Development Agreement (LP-CRADA) with the Government. Submitting an application in response to this RFI does not guarantee an invitation to participate in the experiment.

4.0 Review of Vendor Responses

Responses to this RFI may be reviewed by Government technical experts drawn from staff within DHS and other Federal agencies. The Government will review vendor responses for potential participation in the experiment. The Government does not intend to provide a response to submissions for this RFI, but the Government reserves the right to meet with respondents, either individually or as a group, regarding this RFI. These meetings may be scheduled at the Government's discretion but are not guaranteed.

4 1.1 **Special notice**

While only U.S. or Canadian Government personnel will participate in the final selection process, Government support contractors may review and provide support during the application evaluation process. All participants involved in the evaluation process, including those in support roles, will sign a Non-Disclosure Agreement prior to such involvement. Submission in response to this RFI constitutes approval to release the applications to the evaluation team that may include Government support contractors.